SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

RP-Cloud Labs - Tool-Session Service

Course: ITLSE801 - Software Engineering

Project: RP-Cloud Labs

Document Type: Mini Software Requirements Specification

Version: 1.0

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of the RP-Cloud Labs Tool-Session Service. The document provides a foundation for software development, testing, and evaluation.

1.2 Scope

The Tool-Session Service will allow students to request access to hosted learning tools through a browser. It will validate student accounts, check available college software license capacity, process session requests, and record approved sessions.

The service aims to improve access to learning tools and provide a consistent software environment for practical learning across Rwanda Polytechnic colleges.

1.3 Definitions

- **Tool session:** A period during which a student is granted access to a hosted learning tool.
- **License capacity:** The number of sessions permitted by the available software licenses.
- **Stakeholder:** A person or organization that uses, manages, develops, or is affected by the system.
- **MVP:** Minimum Viable Product, the initial version contains the essential features.
- **SRS:** Software Requirements Specification.

2. Overall Description

2.1 Product Perspective

The Tool-Session Service is a component of the larger RP-Cloud Labs platform. It will work with authentication services, college information, license-capacity tracking, session storage, and hosted learning tools.

2.2 User Classes

1. **Students:** Select learning tools, submit session requests, and view request outcomes.
2. **Lecturers:** Use hosted learning tools to support teaching and practical assessments.
3. **College administrators:** Support college onboarding and maintain college-related information.
4. **RP central IT administrators:** Manage access, monitor service performance, and maintain security.
5. **Development team:** Implement, test, document, and maintain the service.

2.3 Assumptions and Constraints

The system assumes that reliable student account information and current license-capacity information are available. Access to hosted tools depends on the availability of the required infrastructure and valid software licenses.

The development team must work within the agreed project scope, budget, timeline, and technology constraints.

3. Functional Requirements

FR-01: Submit Tool-Session Request

The system shall allow a student to submit a tool-session request by selecting an available learning tool and submitting the request.

Acceptance criteria:

- The student can select an available learning tool.
- The system receives the submitted request.
- The system provides a clear response indicating the request outcome.

FR-02: Validate Student Account

The system shall reject a tool-session request if the student's account has not been validated.

Acceptance criteria:

- Validated accounts can proceed to the license-capacity check.
- Unvalidated accounts cannot obtain a session through the service.
- The system provides an appropriate rejection message.

FR-03: Check College License Capacity

The system approve a tool-session request only when the student's account is validated and sufficient license capacity is available for the selected tool at the student's college.

Acceptance criteria:

- The system checks the capacity associated with the selected tool and college.
- A request is approved only when all required conditions are satisfied.
- If no capacity is available, the system does not approve the request and informs the student.

FR-04: Record Approved Session

The system shall record every approved session with a unique session ID, student ID, college ID, tool ID, start time, and session status.

Acceptance criteria:

- Every approved session has a unique identifier.
- All required fields are saved.

- The stored session record can be retrieved for verification.

4. Non-Functional Requirements (FURPS+)

NFR-01- Functionality: The system shall enforce student account-validation and license-capacity rules for every tool-session request.

NFR-02 - Usability: After signing in, a student shall be able to submit a tool-session request through a browser using no more than five user actions.

NFR-03 - Reliability: The system shall preserve successfully approved session records when the service is restarted.

NFR-04 - Performance: The session-request API shall return a response within two seconds for at least 95% of requests under a test load of 100 concurrent requests, excluding external tool startup and streaming time.

NFR-05 - Supportability: The system shall record errors and failed session requests with a timestamp, request identifier, and error category in its application logs.

NFR-06 - Security (+): The system shall deny access to protected session-management operations unless the requester is authenticated and authorized.

These requirements will be verified through usability testing, functional tests, restart tests, load testing, log inspection, and access-control testing. The proposed numerical performance and usability targets must be confirmed by the project team before implementation.

5. External Interface Requirements

5.1 User Interface

The system should provide a browser-based interface that allows students to select available learning tools, submit session requests, and view request outcomes. The interface shall display clear messages for approved, rejected, or unsuccessful requests.

5.2 Software Interfaces

The service shall communicate with the relevant authentication component, college information source, license-capacity tracking mechanism, and database for storing session records.

Redis may be used for shared capacity tracking and Apache Kafka for communicating session events between services.

5.3 Hardware and Network Interfaces

Students should access the service using an internet-connected device and a supported web browser. Hosted learning tools shall execute on server infrastructure rather than requiring students to install the software locally.

5.4 Communication Interfaces

The user interface shall communicate with the service through defined application endpoints over HTTP or HTTPS. HTTPS shall be used in deployment to protect communication in transit.

Conclusion

This mini-SRS defines the purpose, users, functional requirements, quality requirements, and external interfaces of the RP-Cloud Labs Tool-Session Service. It will guide implementation and testing of the initial MVP and provide a basis for future improvements.

Milestone Link: https://github.com/fredndabikunze01/rp-cloud-labs-tool-session-service/milestone/1